Qualitative and Quantitative Performance Evaluation of U-Net and U-Net++ for Road Segmentation using KITTI Dataset

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Abstract

The precise identification of road areas within autonomous driving systems is a critical component in trajectory planning and ensuring safety. This study presents a comparative analysis of the road segmentation performance of U-Net, a quintessential semantic segmentation model, and its advanced variant, U-Net++, using the KITTI Road Dataset.

To ensure a rigorous comparison, identical preprocessing protocols and data augmentation techniques via the Albumentations library were applied to both architectures. Training was conducted employing a composite loss function incorporating Binary Cross Entropy and the Dice Coefficient. Quantitative performance assessment was performed on the basis of the Intersection over Union (IoU) metric.

Experimental results reveal that the baseline U-Net model achieved an IoU score of approximately 0.8530, whereas the U-Net++ model attained a superior IoU score of approximately 0.8779. These findings demonstrate that U-Net++ effectively bridges the semantic gap between the encoder and decoder through dense skip paths and deep supervision, thus exhibiting enhanced efficacy in delineating road boundaries and fine-grained regions. Consequently, this study quantitatively validates that U-Net++ provides more precise segmentation capabilities compared to conventional U-Net in the context of road segmentation for autonomous driving applications.

1 Introduction

Introduction Semantic scene understanding is a key element of advanced driving assistance systems (ADAS) and autonomous driving (AD) [1]. To achieve precise road segmentation, deep learning models have evolved significantly. As demonstrated in foundational studies, convolutional networks by themselves, trained end-to-end, pixels-to-pixels, exceed the state-of-the-art in semantic segmentation [2].

Currently, the state-of-the-art models for image segmentation are variants of the encoder-decoder architecture like U-Net [3]. U-Net utilizes skip connections to recover spatial information; however, it often struggles with the semantic disparity between encoder and decoder features. To address this, U-Net++ introduces a novel architecture where the re-designed skip pathways aim at reducing the semantic gap between the feature maps of the encoder and decoder sub-networks [3].

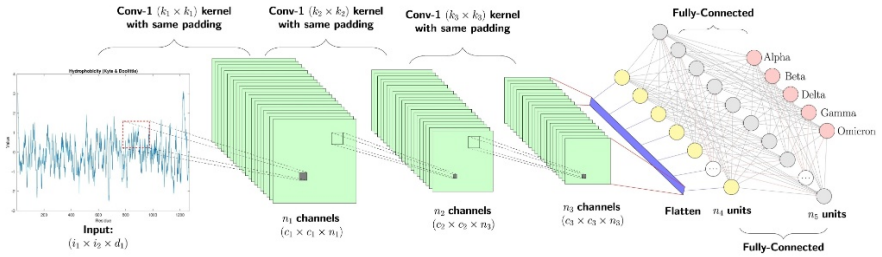


Figure 1: The architecture of U-Net, illustrating the contracting path (encoder) and expansive path (decoder) with skip connections.

This paper presents a comparative study of U-Net and U-Net++ on the KITTI dataset. By evaluating their performance quantitatively and qualitatively, we verify the efficacy of reducing the semantic gap for robust road segmentation in autonomous driving scenarios.

2 Msthod

2.1 Dataset and Preprocessing

1. Dataset Configuration

   To evaluate the road segmentation performance in autonomous driving environments, we utilized the KITTI Road Dataset. This dataset comprises high-resolution images captured in diverse urban scenarios (e.g., Urban marked, Urban unmarked, Urban multiple marked lanes) along with their corresponding pixel-level Ground Truth (GT) labels.For the experiments, the entire dataset was randomly shuffled and partitioned into a training set and a validation set with a ratio of 8:2. The validation set was employed to monitor the training progress and prevent overfitting. To ensure memory efficiency and standardize the input for the models, all original images were resized to 224×224 pixels. Furthermore, to facilitate rapid convergence, a normalization process was applied by scaling the pixel values of all input images to the range of $[0, 1]$.

2. Data Augmentation via Albumentations.

Deep learning models, particularly encoder-decoder architectures like U-Net, are prone to overfitting when trained on limited datasets. To mitigate this issue and enhance the generalization capability of the model, we employed data augmentation techniques using the Albumentations library during the training phase.

The primary augmentation strategies implemented in this study are as follows.

- Horizontal Flip: Applied with a probability of 50% to flip the image horizontally, enabling the model to learn robustness against directional changes in the road environment.
- Random Sized Crop: A specific region of the original image is randomly cropped and then resized back to 224×224 pixels. This technique allows the model to learn road features across varying scales.
- Resize: All processed images are finally resized to a fixed resolution of 224×224 pixels to maintain consistency in the input dimension.

It is important to note that these augmentation techniques were exclusively applied to the training set. For the validation and testing phases, only standard resizing and normalization were performed to ensure a consistent and fair performance evaluation.

2.2 Model Architecture and Training Strategy

1. Comparison of U-Net and U-Net++ Architectures

   In this study, we conducted a comparative analysis of U-Net and U-Net++, both of which are based on the encoder-decoder architecture, for the task of semantic segmentation.

   - U-Net

     This model features a U-shaped architecture consisting of a symmetric contracting path (encoder) and an expansive path (decoder). It utilizes skip connections to concatenate feature maps from the encoder to the corresponding layers in the decoder, thereby facilitating the recovery of spatial information. However, this simple concatenation has a limitation in that it does not fully resolve the semantic gap between the low-level features of the encoder and the high-level features of the decoder.

```
UNet(
  (enc1): Sequential(
    (0): Conv2d(3, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU(inplace=True)
    (2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
```

```
    (3): ReLU(inplace=True)
  )
  (pool1): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
    ceil_mode=False)
  (enc2): Sequential(
    (0): Conv2d(64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU(inplace=True)
    (2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (3): ReLU(inplace=True)
  )
  (pool2): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
    ceil_mode=False)
  (enc3): Sequential(
    (0): Conv2d(128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU(inplace=True)
    (2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (3): ReLU(inplace=True)
  )
  (pool3): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
    ceil_mode=False)
  (enc4): Sequential(
    (0): Conv2d(256, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU(inplace=True)
...
    (2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (3): ReLU(inplace=True)
  )
  (final): Conv2d(64, 1, kernel_size=(1, 1), stride=(1, 1))
)
```

Listing 1: UNet Architecture

- U-Net++

  Proposed to address the limitations of U-Net, U-Net++ introduces nested dense skip pathways. By redesigning the skip connections to include dense convolution blocks between the encoder and decoder, the model gradually enriches the feature maps with semantic information before they are fused in the decoder. This mechanism effectively bridges the semantic gap, enabling more precise segmentation of fine-grained details such as road boundaries.

```
UNetPlusPlus(
  (pool): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode
    =False)
  (up): Upsample(scale_factor=2.0, mode='bilinear')
  (conv0_0): Sequential(
    (0): Conv2d(3, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU(inplace=True)
    (2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (3): ReLU(inplace=True)
  )
  (conv1_0): Sequential(
    (0): Conv2d(64, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU(inplace=True)
    (2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (3): ReLU(inplace=True)
  )
  (conv2_0): Sequential(
    (0): Conv2d(128, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU(inplace=True)
    (2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (3): ReLU(inplace=True)
  )
  (conv3_0): Sequential(
    (0): Conv2d(256, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU(inplace=True)
    (2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
...
    (2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (3): ReLU(inplace=True)
  )
```

```
  (final): Conv2d(64, 1, kernel_size=(1, 1), stride=(1, 1))
)
```

Listing 2: UNet++ Architecture

2. Loss Function

   Road segmentation can be formulated as a pixel-wise binary classification problem. To mitigate the class imbalance issue and maximize segmentation performance, we employed a composite loss function that combines Binary Cross Entropy (BCE) Loss and Dice Loss.

$$L_{Total} = L_{BCE} + L_{Dice}$$

- Binary Cross Entropy (BCE) Loss
  This loss measures the distributional difference between the predicted probability and the actual label for each pixel, thereby enhancing pixel-level classification accuracy.

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$$

- Dice Loss
  This metric quantifies the overlap between the predicted region and the ground truth. It is particularly effective in ensuring stable training even when the road area (foreground) is significantly smaller than the background.

$$L_{Dice} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i + \epsilon}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i + \epsilon}$$

3 Results and Analysis

3.1 Experimental Setup and Evaluation Metrics

1. Implementation Details and Hyperparameters

   All experiments in this study were implemented using the PyTorch deep learning framework. To ensure a fair comparison, identical hyperparameter settings were applied to both the U-Net and U-Net++ models.

   The training process was conducted for 100 epochs, utilizing the Adam optimizer for parameter optimization. The learning rate and batch size were tuned to optimal values, considering GPU memory efficiency and stable convergence. Furthermore, to mitigate overfitting, we adopted a strategy of early stopping or learning rate adjustment if the validation loss did not improve over a certain number of epochs.

2. Evaluation Metrics

   o quantitatively evaluate the road segmentation performance, we employed the Intersection over Union (IoU) metric, which is a standard benchmark in the field of semantic segmentation. The IoU measures the overlap between the predicted segmentation mask and the ground truth, calculated as the ratio of the area of intersection to the area of union.

   In this study, the IoU for the road class is defined as follows.

$$IoU = \frac{|A \cap B|}{|A \cup B|} = \frac{TP}{TP + FP + FN}$$

   Where A denotes the predicted road region and B denotes the ground truth road region. The metric can also be expressed in terms of the confusion matrix components.

- TP (True Positive): The number of pixels correctly predicted as road.

Models	IoU Score
UNet	0.852968
UNet++	0.877923

Table 1: IoU Score by Models



Figure 2: Visualization of UNet prediction results

- FP (False Positive): The number of background pixels incorrectly predicted as road.
- FN (False Negative): The number of road pixels incorrectly predicted as background.

The IoU metric provides an objective assessment of segmentation performance, particularly in scenarios involving class imbalance where the background area significantly dominates the road area.

3.2 Evaluation Result

1. Quantitative Results

   The performance of U-Net and U-Net++ on the KITTI road segmentation task was evaluated using the test set. The quantitative results are summarized in Table 1.

2. Qualitative Results

   To visually assess the segmentation quality, we selected challenging test samples, such as scenes with shadows or curved roads.

   - Figure 2 and Figure 3 present a visual comparison of the segmentation performance, illustrating the predicted results from the U-Net and U-Net++ models, respectively.
   - Visual Analysis

     While U-Net successfully captured the overall road structure, it exhibited limitations in delineating sharp boundaries and produced false negatives in shadowed regions. In contrast, U-Net++ generated smoother and more accurate boundary delineations. Notably, it



Figure 3: Visualization of UNet++ prediction results

demonstrated robustness against illumination changes caused by shadows, producing segmentation masks that closely resemble the Ground Truth.

References

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[2] Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3431–3440, 2015.

[3] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. Unet++: A nested u-net architecture for medical image segmentation. In *International workshop on deep learning in medical image analysis*, pages 3–11. Springer, 2018.